CSF3123

FitLife GYM Membership Management System

Group 11

- NUR AYUNI BINTI AZIZI S76255
- NURUL DYAASYKIN BINTI MOHD EZANY S76282
- NUR SYUHADAH BINTI ZAIMI S74290
- DANISH HATEEM BIN JUMAT S74742

Introduction

FitLife Gym focuses on meeting client needs through efficient membership registration and quality services. However, many of its operations still rely on paper-based records and separate digital tools, which causes inefficiencies and data inconsistencies in managing members, trainers, and payments.

Currently, member details, trainer information, and payment records are handled manually. This often leads to missing records, billing errors, and confusion among members regarding their trainers and payment status, while staff spend extra time correcting these issues.

This project proposes a centralized gym membership management system using a relational database. All information will be stored in one system, allowing members to view their personal details, trainer information, and payment status, while staff access data based on their roles.

Overall ,the system reduces manual data checking, improves accuracy, and enhances service efficiency using an online database.



Problem Statement

1 Inefficient Client & Trainer Information Management

- Paper-based records
- Data duplication

- Inconsistent information
- Extra administrative work

- Centralized online database
- Structured client & trainer tables

2 Manual Membership & Payment Tracking

- Manual payment records
- No centralized tracking

- Missing or outdated payment data
- Client confusion on payment status

- Online payment records
- Clients view payment status & last payment date

3 Lack of Controlled Data Access

- No access restrictions

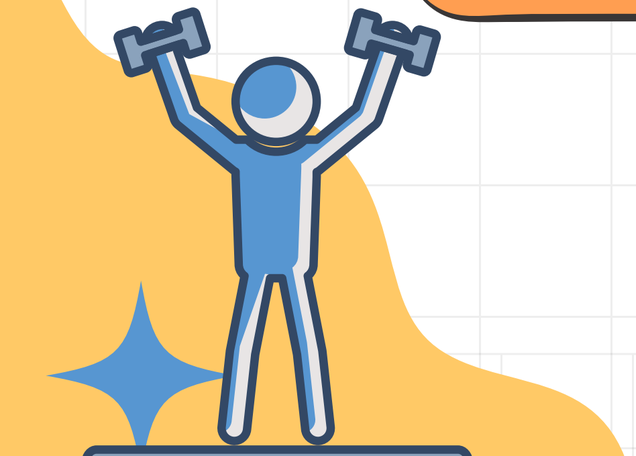
- Unauthorized data access
- Security risks

- Role-based access control
- Limited access by user role

Goals and Objective

- Provide a centralized system for storing and managing
- Improve operational efficiency
- Reduce errors associated with manual data entry and support better decision-making
- Enable automatic generation of reports

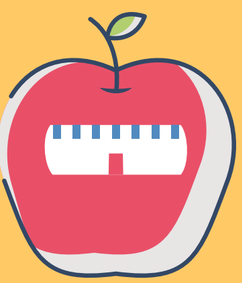
OBJECTIVE

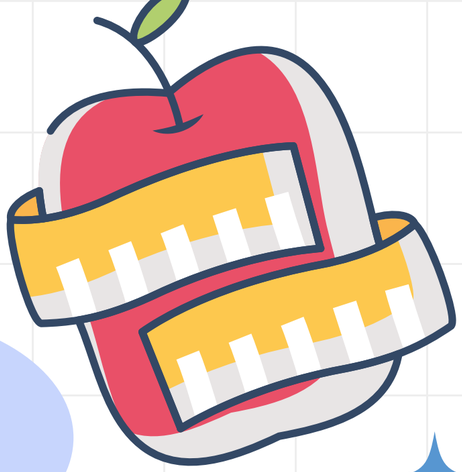


GOAL

- CILENT / USER GOALS
  - STREAMLINE MEMBER ONBOARDING
  - STANDARDIZE PROGRESS TRACKING
  - FINANCIAL OPERATIONS
- CLIENTS / USER REQUIREMENTS
  - MEMBER INFORMATION ACCESS
  - TRAINER INFORMATION VISIBILITY
- DATABASE GOALS
  - CENTRALIZATION OF OPERATIONS
  - USER-FRIENDLY DESIGN
  - ACCURATE PAYMENT TRACKING

User Profile





Admin

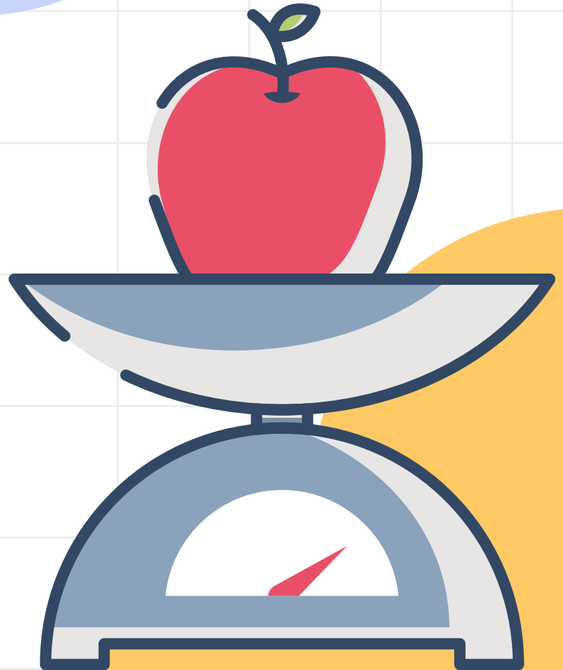
- Full access to the system
- Registers customers and trainers
- Updates membership details, records payments, and generates reports

Customer

- Gym members using the system
- Personal and membership details stored
- Limited access; cannot modify sensitive data
- Simple and user-friendly interface

Trainer

- Provides fitness training
- Trainer details stored (name, contact, expertise)
- Can view customer training information only and not their payment status



Waterfall Method

REQUIREMENT ANALYSIS

- System problems and objectives
- Determine user roles
- System requirements and business rules

SYSTEM & DATABASE DESIGN

- Design the ERD
- Define table relationships
- Normalisation

IMPLEMENTATION

- Create database tables & Insert data
- Develop SQL queries (CRUD operations)
- Design user interface mockups

TESTING & VALIDATION

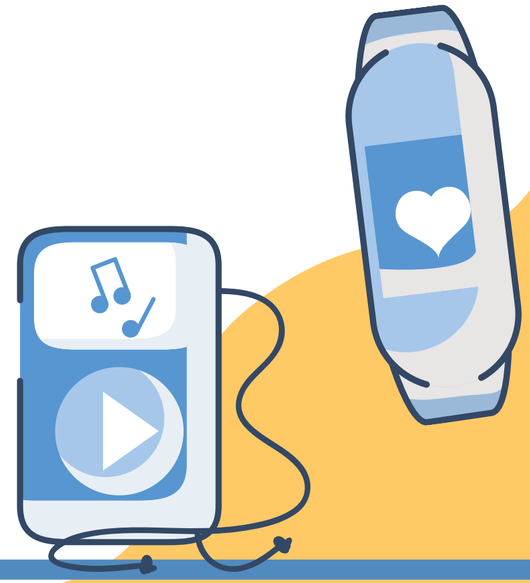
- Test system functionality
- Ensure requirements are met
- Validate data accuracy and role access

DEPLOYMENT & EVALUATION

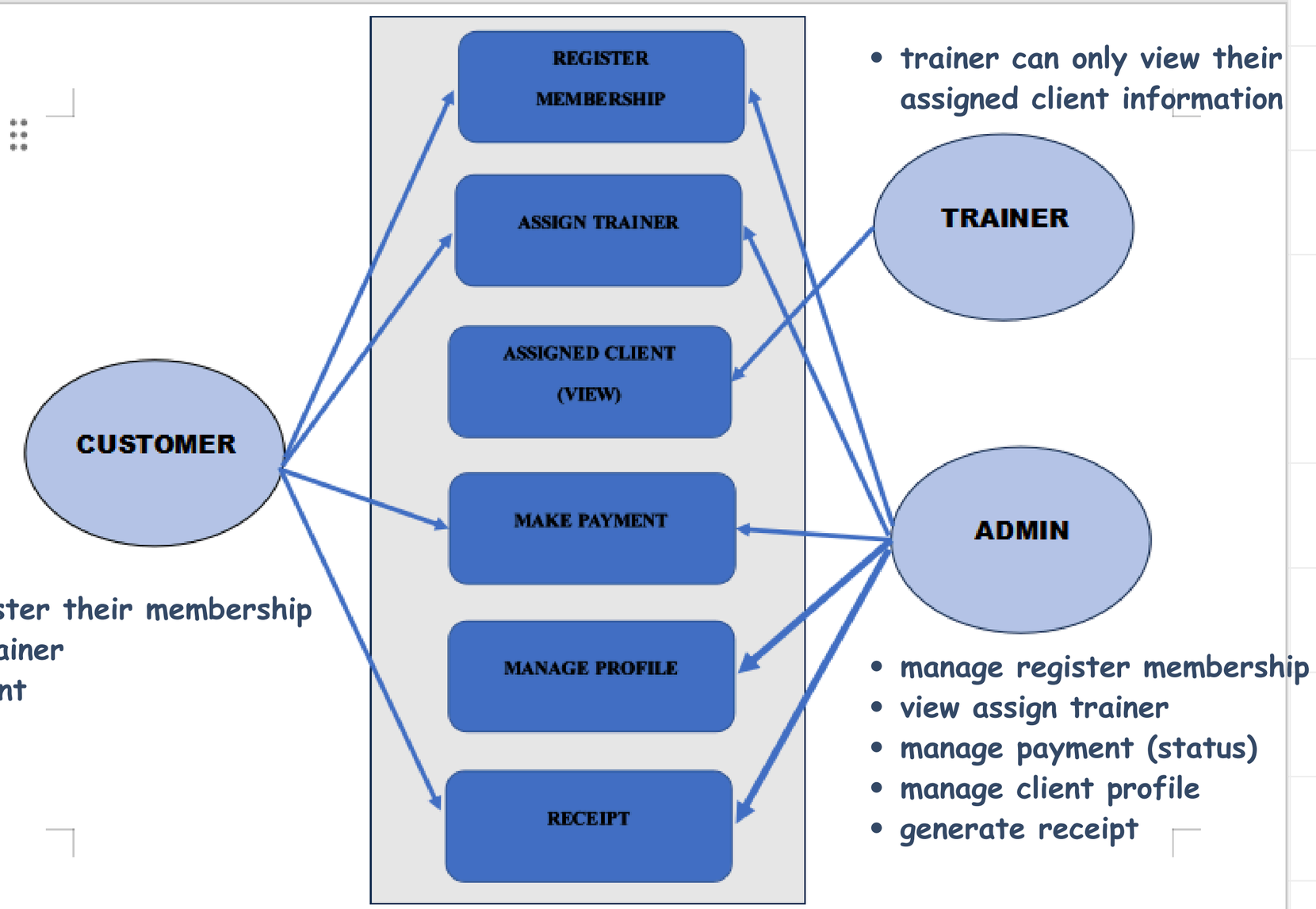
- Review system performance
- System strengths and weaknesses

MAINTANANCE

- Suggest improvements
- Future system upgrades



Use Case Diagram



- trainer can only view their assigned client information

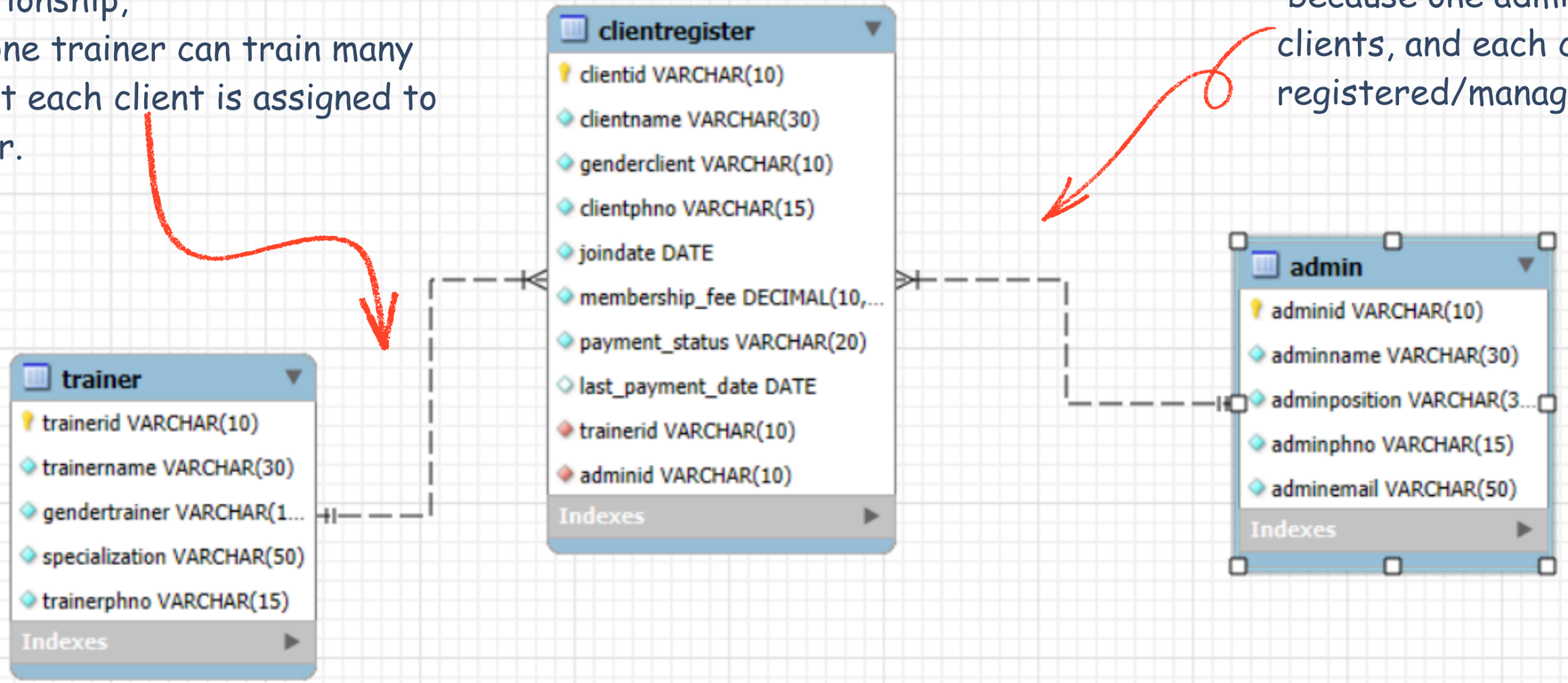
- customer register their membership
- view assign trainer
- make a payment
- view receipt

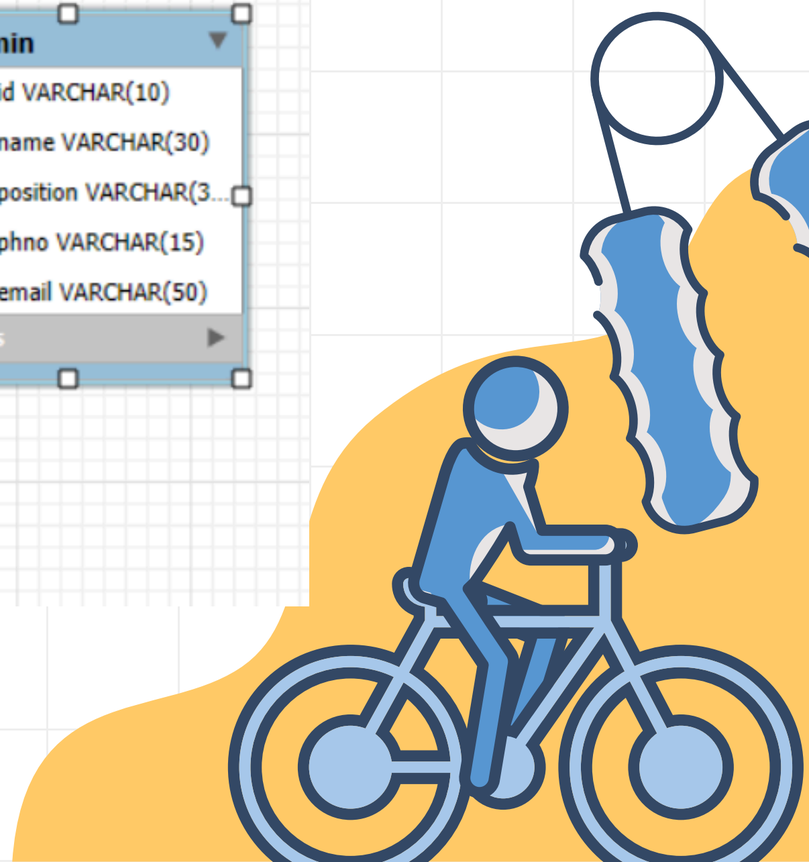
- manage register membership
- view assign trainer
- manage payment (status)
- manage client profile
- generate receipt

ERD

Trainer and Client have a one-to-many (1:M) relationship,
because one trainer can train many clients, but each client is assigned to one trainer.

Admin and Client have a many-to-one (M:1) relationship,
because one admin can manage many clients, and each client is registered/managed by one admin.





SQL Query

In this project, we use the CRUD concept, which stands for Create, Retrieve, Update, and Delete.

Create

```sql
CREATE DATABASE IF NOT EXISTS gym;
USE gym;

CREATE TABLE admin (
    adminid VARCHAR(10) NOT NULL,
    adminname VARCHAR(30) NOT NULL,
    adminposition VARCHAR(30) NOT NULL,
    adminphno VARCHAR(15) NOT NULL,
    adminemail VARCHAR(50) NOT NULL,
    CONSTRAINT admin_adminid_pk PRIMARY KEY (adminid)
);

CREATE TABLE trainer (
    trainerid VARCHAR(10) NOT NULL,
    trainername VARCHAR(30) NOT NULL,
    gendertrainer VARCHAR(10) NOT NULL CHECK (gendertrainer IN ('Male', 'Female', 'Other')),
    specialization VARCHAR(50) NOT NULL,
    trainerphno VARCHAR(15) NOT NULL,
    CONSTRAINT trainer_trainerid_pk PRIMARY KEY (trainerid)
);

CREATE TABLE clientregister (
    clientid VARCHAR(10) NOT NULL,
    clientname VARCHAR(30) NOT NULL,
    genderclient VARCHAR(10) NOT NULL CHECK (genderclient IN ('Male', 'Female', 'Other')),
    clientphno VARCHAR(15) NOT NULL,
    joindate DATE NOT NULL,
    membership_fee DECIMAL(10,2) NOT NULL CHECK (membership_fee > 0),
    payment_status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    last_payment_date DATE,
    trainerid VARCHAR(10) NOT NULL,
    adminid VARCHAR(10) NOT NULL,
    CONSTRAINT clientregister_clientid_pk PRIMARY KEY (clientid),
    CONSTRAINT clientregister_trainerid_fk FOREIGN KEY (trainerid) REFERENCES trainer(trainerid),
    CONSTRAINT clientregister_adminid_fk FOREIGN KEY (adminid) REFERENCES admin(adminid)
);
```

Create View (Client Profile)

```sql
CREATE VIEW client_view AS
SELECT
    c.clientid          AS client_id,
    c.clientname        AS client_name,
    c.genderclient      AS gender,
    c.clientphno        AS phone,
    c.joindate          AS join_date,
    c.membership_fee    AS membership_fee,
    c.payment_status    AS payment_status,
    c.last_payment_date AS last_payment,
    t.trainername       AS trainer_name,
    t.specialization    AS trainer_specialization,
    t.trainerphno       AS trainer_contact
FROM clientregister c
JOIN trainer t
    ON c.trainerid = t.trainerid;

SELECT *
FROM client_view
WHERE client_id = 'C001';
```

client_id	client_name	gender	phone	join_date	membership_fee	payment_status	last_payment	trainer_name	trainer_specialization	trainer_contact
C001	Alice Wong	Female	60176543210	2024-01-15	150.00	Paid	2025-01-15	Mike Johnson	Weight Training	60198765432



SQL Query

Create Join Table (clientdata+trainername)

```
2 •  SELECT c.clientid, c.clientname, c.membership_fee, c.payment_status,
3            c.last_payment_date, t.trainername, a.adminname
4    FROM clientregister c
5    JOIN trainer t ON c.trainerid = t.trainerid
6    JOIN admin a ON c.adminid = a.adminid;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

clientid	clientname	membership_fee	payment_status	last_payment_date	trainername	adminname
C001	Alice Wong	150.00	Paid	2025-01-15	Mike Johnson	John Smith
C002	Bob Tan	150.00	Paid	2025-02-20	Sarah Lee	John Smith
C005	Eva Chen	150.00	Overdue	2025-04-01	Mike Johnson	John Smith
C003	Charlie Lim	150.00	Pending	2025-03-10	Mike Johnson	Mary Johnson
C004	Diana Lee	200.00	Paid	2025-03-15	David Chen	Mary Johnson

Delete(client by specific id)

```
1 •  DELETE FROM clientregister
2    WHERE clientid = 'C001';
3
4 •  SELECT * FROM clientregister
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

clientid	clientname	genderclient	clientphno	joindate	membership_fee	payment_status	last_payment_date	trainerid	adminid
C002	Bob Tan	Male	60165432109	2024-02-20	150.00	Paid	2025-02-20	T002	A001
C003	Charlie Lim	Male	60154321098	2024-03-10	150.00	Paid	2026-01-02	T001	A002
C004	Diana Lee	Female	60143210987	2024-03-15	200.00	Paid	2025-03-15	T003	A002
C005	Eva Chen	Female	60132109876	2024-04-01	150.00	Overdue	2025-04-01	T001	A001
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL





SQL Query

Retrieve (client table)

```
1 •  SELECT * FROM clientregister
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

clientid	clientname	genderclient	clientphno	joindate	membership_fee	payment_status	last_payment_date	trainerid	adminid
C001	Alice Wong	Female	60176543210	2024-01-15	150.00	Paid	2025-01-15	T001	A001
C002	Bob Tan	Male	60165432109	2024-02-20	150.00	Paid	2025-02-20	T002	A001
C003	Charlie Lim	Male	60154321098	2024-03-10	150.00	Pending	2025-03-10	T001	A002
C004	Diana Lee	Female	60143210987	2024-03-15	200.00	Paid	2025-03-15	T003	A002
C005	Eva Chen	Female	60132109876	2024-04-01	150.00	Overdue	2025-04-01	T001	A001
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Retrieve payment status

```
2 •  SELECT clientid, clientname, clientphno, membership_fee, payment_status
3    FROM clientregister
4    WHERE payment_status IN ('Pending', 'Overdue');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

clientid	clientname	clientphno	membership_fee	payment_status
C003	Charlie Lim	60154321098	150.00	Pending
C005	Eva Chen	60132109876	150.00	Overdue
NULL	NULL	NULL	NULL	NULL

Update(specific client's phone number)

```
1 •  UPDATE clientregister
2    SET clientphno = '60176543212'
3    WHERE clientid = 'C001';
4
5 •  SELECT * FROM clientregister
6    WHERE clientid = 'C001' ;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

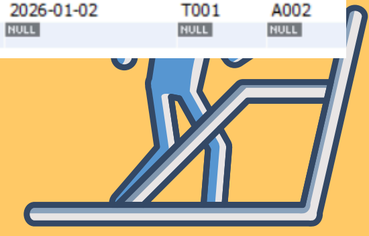
clientid	clientname	genderclient	clientphno	joindate	membership_fee	payment_status	last_payment_date	trainerid	adminid
C001	Alice Wong	Female	60176543212	2024-01-15	150.00	Paid	2025-01-15	T001	A001
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Update(specific client's payment status)

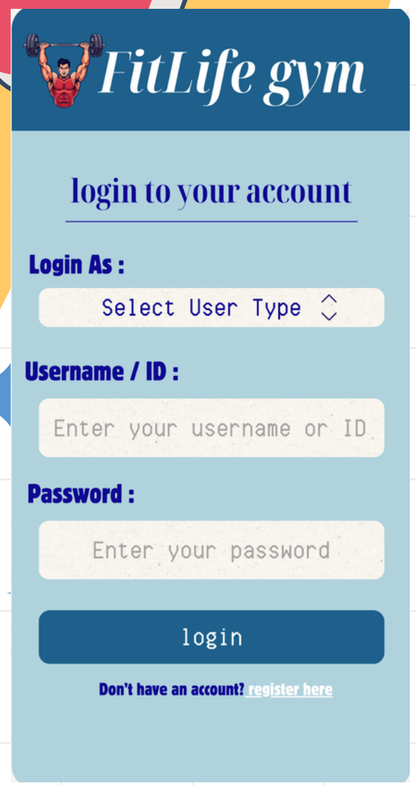
```
2 •  UPDATE clientregister
3    SET payment_status = 'Paid', last_payment_date = '2026-01-02'
4    WHERE clientid = 'C003';
5
6 •  SELECT * FROM clientregister
7    WHERE clientid = 'C003' ;
```

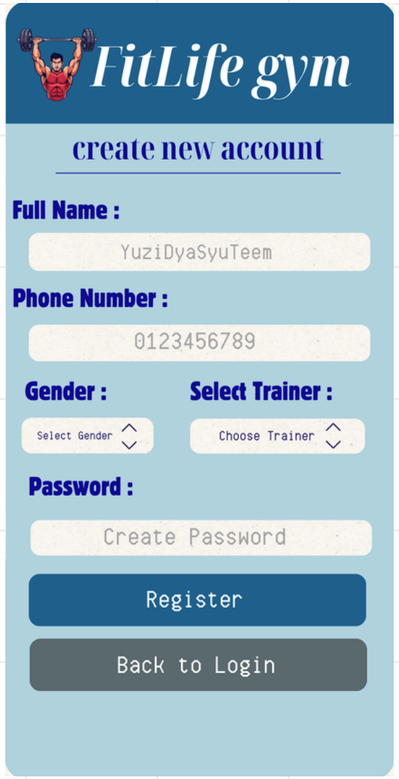
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

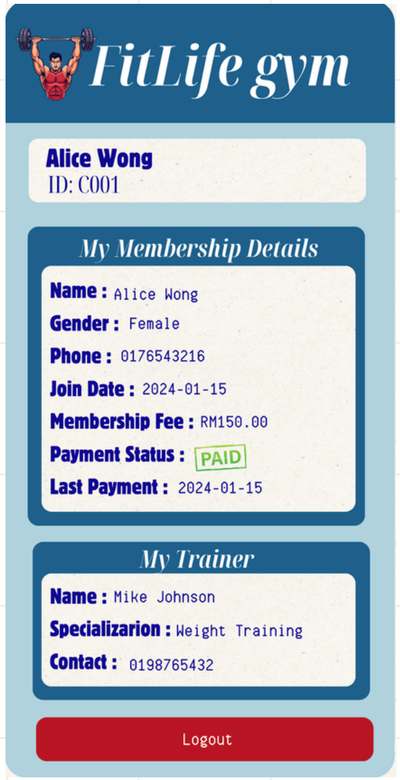
clientid	clientname	genderclient	clientphno	joindate	membership_fee	payment_status	last_payment_date	trainerid	adminid
C003	Charlie Lim	Male	60154321098	2024-03-10	150.00	Paid	2026-01-02	T001	A002
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

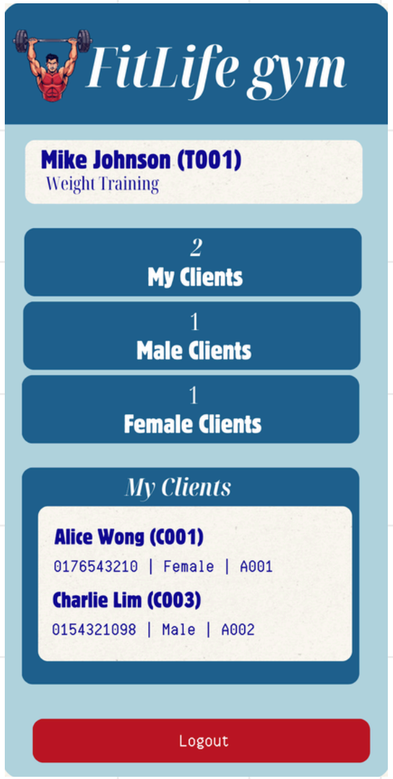


Interface


FitLife gym
login to your account
Login As :
Select User Type
Username / ID :
Enter your username or ID
Password :
Enter your password
login
Don't have an account? register here


FitLife gym
create new account
Full Name :
YuziDyaSyuTeem
Phone Number :
0123456789
Gender :
Select Gender
Select Trainer :
Choose Trainer
Password :
Create Password
Register
Back to Login


FitLife gym
Alice Wong
ID: C001
My Membership Details
Name : Alice Wong
Gender : Female
Phone : 0176543216
Join Date : 2024-01-15
Membership Fee : RM150.00
Payment Status : PAID
Last Payment : 2024-01-15
My Trainer
Name : Mike Johnson
Specializarion : Weight Training
Contact : 0198765432
Logout


FitLife gym
Mike Johnson (T001)
Weight Training
2
My Clients
1
Male Clients
1
Female Clients
My Clients
Alice Wong (C001)
0176543210 | Female | A001
Charlie Lim (C003)
0154321098 | Male | A002
Logout


FitLife gym
John Smith (A001)
Manager
3
Total Clients
RM 300.00
Total Revenue
1
Pending Payments
3
Active Trainers
Client Details
Logout


FitLife gym
clients details
Edit
CLIENTS DETAILS
C001
C002
C003
NAME
Alice Wong
Bob Tan
Charlie Lim
GENDER
Female
Male
Male
PHONE
0176543210
0165432109
0154321098
JOINED DATE
2024-01-15
2024-02-20
2024-03-10
MEMBERSHIP FEE
150.00
150.00
150.00
PAYMENT STATUS
Paid
Paid
Pending
LAST PAYMENT
2025-01-15
2025-02-20
2025-03-10
TRAINER NAME
MIKE JOHNSON
SARAH LEE
MIKE JOHNSON
ADMIN NAME
JOHN SMITH
JOHN SMITH
MARRY JOHNSON
Back

That's all from us ,
Thank You !



